Matrix diagnolization example step by step problem solving in details

Let's go through matrix diagonalization step by step with a clear example.

🔷 Example Problem

Diagonalize the matrix:

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

We want to find:

$$A = PDP^{-1}$$

Where:

- D = diagonal matrix (eigenvalues)
- P = matrix of eigenvectors

✅ Step 1: Find Eigenvalues

Use the characteristic equation:

$$\det(A - \lambda I) = 0$$

$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)(3-\lambda) - 2 = 0$$

Expand:

$$(4-\lambda)(3-\lambda) - 2 = 0$$

$$12 - 4\lambda - 3\lambda + \lambda^2 - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 5)(\lambda - 2) = 0$$

✔ Eigenvalues:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

✅ Step 2: Find Eigenvectors

🔹 For $\lambda = 5$

Solve:

$$(A - 5I)X = 0$$

$$\begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix}$$

Equation:

$$-x + y = 0 \Rightarrow y = x$$

✔ Eigenvector:

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

🔹 For $\lambda = 2$

$$(A - 2I) = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix}$$

Equation:

$$2x + y = 0 \Rightarrow y = -2x$$

✔ Eigenvector:

$$v_2 = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

✅ Step 3: Form Matrix P and D

Matrix P (eigenvectors as columns):

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}$$

Matrix D:

$$D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

✅ Step 4: Find P^{-1}

$$P^{-1} = \frac{1}{(1)(-2) - (1)(1)} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix}$$

$$= \frac{1}{-3} \begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix}$$

🎯 Final Answer

$$A = PDP^{-1}$$

Where:

$$P = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix}, \quad D = \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix}$$

🔥 Key Idea (Very Important)

Matrix is diagonalizable only if:

- It has enough independent eigenvectors

💡 Quick Summary

1. Find eigenvalues
2. Find eigenvectors
3. Form P using eigenvectors
4. Form D using eigenvalues
5. Verify: $A = PDP^{-1}$

If you want, I can give:

✅ ↳ 3×3 diagonalization example

✅ ↳ Non-diagonalizable case (very important for exams)

✅ ↳ Short tricks to solve faster

Just tell me 👍

How we know that vectors are independent or dependent

Good question — this is very important for diagonalization because everything depends on whether eigenvectors are independent.

Let's make it simple and clear.

🔷 What does "independent vectors" mean?

Vectors are linearly independent if:

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$$

has only one solution:

$$c_1 = c_2 = \cdots = 0$$

$$c_1v_1 + c_2v_2 + \cdots + c_nv_n = 0$$

If there is any non-zero solution, then vectors are dependent.

✅ Simple Intuition (Very Easy)

✔ Independent:

- No vector is a multiple of another
- Each vector gives new direction

❌ Dependent:

- One vector = multiple of another
- Or can be formed using others

🔷 Method I: Check Using Multiples (Fast Method)

Example:

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}$$

Here:

$$v_2 = 2 \cdot v_1$$

❌ Dependent

Example:

$$v_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Not multiples → ✔ Independent

🔷 Method 2: Determinant Test (Best for 2 or 3 vectors)

Make matrix using vectors as columns:

$$\begin{bmatrix} 1 & 1 \\ 2 & -1 \end{bmatrix}$$

Check determinant:

$$= (1)(-1) - (1)(2) = -3 \neq 0$$

✔ Independent

👉 Rule:

- det ≠ 0 → Independent
- det = 0 → Dependent

🔷 Method 3: Row Reduction (General Method)

Put vectors into matrix and do Gaussian elimination:

If you get a zero row → dependent

🔷 In Diagonalization (Important Shortcut)

You usually don't need full checking!

✔ Rule:

- Different eigenvalues → eigenvectors are ALWAYS independent

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Example:

If eigenvalues are:

$$\lambda_1 = 5, \quad \lambda_2 = 2$$

Then eigenvectors are automatically ✔ independent

⚠️ Important Case

If eigenvalues are same (repeated):

- Then you must check independence manually
- Otherwise matrix may not be diagonalizable

🎯 Final Summary

Case	Result
Vectors are multiples	Dependent ❌
Determinant ≠ 0	Independent ✔
Different eigenvalues	Always independent ✔
Zero row after elimination	Dependent ❌

If you want, I can show:

✔ ↪ Example where eigenvalues same but vectors dependent (not diagonalizable)

✔ ↪ Trick to check independence in 3×3 quickly

Just tell me 👍

If all eigen values are positive then it will look positive definite

Short answer: Yes — but with an important condition.

✅ Main Rule

A matrix is positive definite if and only if:

👉 All eigenvalues are positive
✔ AND the matrix is symmetric (i.e., $A = A^T$)

🔷 Why symmetry matters?

Positive definiteness is defined using:

$$x^T Ax > 0 \quad \text{for all } x \neq 0$$

This property behaves nicely only for symmetric matrices.

🔷 Key Theorem

For a real symmetric matrix:

- All eigenvalues > 0 ⟹ ✔ Positive definite
- Any eigenvalue ≤ 0 ⟹ ❌ Not positive definite

🔥 Example (Works)

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

- Symmetric ✔
- Eigenvalues = 3, 1 (both positive)

👉 ✔ Positive definite

⚠️ Counter Example (Very Important)

$$A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

- Eigenvalues = 1, 1 (both positive) ✔
- BUT not symmetric ❌

👉 This matrix is NOT positive definite

🔷 Quick Decision Rule

Condition	Result
Symmetric + all eigenvalues > 0	✔ Positive definite
Not symmetric (even if eigenvalues > 0)	❌ Not guaranteed
Any eigenvalue ≤ 0	❌ Not positive definite

💡 Easy Way to Remember

👉 "Eigenvalues positive" works only when matrix is nice (symmetric)

🎯 Final Answer

✔ If all eigenvalues are positive AND matrix is symmetric, then it is positive definite

❌ If matrix is not symmetric, then positive eigenvalues do NOT guarantee positive definiteness

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If you want, I can also show:

✔ ↪ Shortcut using principal minors (Sylvester test)

✔ ↪ How to check positive definite without eigenvalues (very useful in exams)

Give example solving

Let's solve a complete example step by step to check whether a matrix is positive definite.

🔷 Example 1 (Positive Definite)

Given:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

✅ Step 1: Check Symmetry

$$A^T = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = A$$

✔ Matrix is symmetric

✅ Step 2: Find Eigenvalues

Solve:

$$\det(A - \lambda I) = 0$$

$$\det \begin{pmatrix} 2-\lambda & 1 \\ 1 & 2-\lambda \end{pmatrix} = 0$$

$$(2-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda - 3)(\lambda - 1) = 0$$

✔ Eigenvalues:

$$\lambda_1 = 3, \quad \lambda_2 = 1$$

✅ Step 3: Conclusion

- Both eigenvalues > 0 ✔
- Matrix is symmetric ✔

👉 A is Positive Definite

🔷 Example 2 (Important – Not Positive Definite)

Given:

$$B = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}$$

✅ Step 1: Check Symmetry

$$B^T = \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} \neq B$$

❌ Not symmetric

✅ Step 2: Eigenvalues

$$\det(B - \lambda I) = (1 - \lambda)^2 = 0$$

✔ Eigenvalues:

$$\lambda = 1, 1 \quad (\text{both positive})$$

⚠️ Step 3: Final Decision

Even though eigenvalues are positive ❗

Matrix is not symmetric

👉 So it is NOT positive definite

🔥 Bonus (Quick Alternative Method – Sylvester Test)

For matrix:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

Check determinants of top-left parts:

- First minor = 2 > 0
- Full determinant = $4 - 1 = 3 > 0$

✔ Both positive → Positive definite

🎯 Final Takeaway

- Symmetric + eigenvalues > 0 → ✔ Positive definite
- Only eigenvalues > 0 → ❌ Not enough
- Use Sylvester test for quick checking

If you want, I can give:

✔ ↪ 3×3 example (very important for exams)

✔ ↪ Case where matrix is negative definite / semi-definite

✔ ↪ Short tricks to solve in 20–30 seconds

Just tell me 👍

Xt AX when snd how we use it any example

Good — this is the core idea behind positive definiteness, so let's make it very clear.

🔷 What is $x^T Ax$?

It is a quadratic form:

$$x^T Ax$$

$x^T Ax$

Where:

- x = column vector
- x^T = row vector (transpose)
- Result is a single number (scalar)

🔷 When do we use $x^T Ax$?

✅ 1. To check Positive Definiteness

👉 Main definition:

$$x^T Ax > 0 \quad \text{for all } x \neq 0$$

✅ 2. In optimization (min/max problems)

Used in:

- Quadratic functions
- Machine learning
- Physics energy equations

✅ 3. In geometry

Represents:

- Length, energy, curvature, etc.

🔷 Step-by-Step Example

Let:

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad x = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

✅ Step 1: Find x^T

$$x^T = [x_1 \quad x_2]$$

✅ Step 2: Multiply Ax

$$Ax = \begin{bmatrix} 2x_1 + x_2 \\ x_1 + 2x_2 \end{bmatrix}$$

✅ Step 3: Multiply $x^T(Ax)$

$$x^T Ax = x_1(2x_1 + x_2) + x_2(x_1 + 2x_2)$$

$$= 2x_1^2 + x_1x_2 + x_1x_2 + 2x_2^2$$

$$= 2x_1^2 + 2x_1x_2 + 2x_2^2$$

✅ Step 4: Analyze

$$= 2(x_1^2 + x_1x_2 + x_2^2)$$

This is always positive (except when $x = 0$)

👉 ✔ So matrix is positive definite

🔥 Try a Value (Very Important Trick)

Take:

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$x^T Ax = 2 + 2 + 2 = 6 > 0$$

Try:

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$= 2 - 2 + 2 = 2 > 0$$

✔ Always positive → confirms

⚠️ Important Insight

If you ever find any x such that:

$$x^T Ax \leq 0$$

👉 Then matrix is NOT positive definite

🎯 Final Summary

Use of $x^T Ax$	Meaning
> 0 for all $x \neq 0$	Positive definite ✔
Sometimes zero	Semi-definite
Negative for some x	Not positive definite ❌

💡 Exam Trick

Instead of checking "all x", just:

- Try 2–3 simple vectors like (1,0), (0,1), (1,1), (1,-1)

If any gives ≤ 0 → ❌ not positive definite

If you want, I can show:

✔ ↪ Example where $x^T Ax = 0$ (semi-definite)

✔ ↪ 3×3 example step-by-step

✔ ↪ Shortcut to avoid long multiplication

Just tell me 👍

Show me a 3×3 example step-by-step

Alright — let's do a full 3×3 example using $x^T Ax$ step by step.

🔷 Example (3×3 Matrix)

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 2 \end{bmatrix}$$

Let:

$$x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

✅ Step 1: Write the quadratic form

We want:

$$x^T Ax$$

$x^T Ax$

✅ Step 2: Find Ax

$$Ax = \begin{bmatrix} 2x_1 - x_2 \\ -x_1 + 2x_2 - x_3 \\ -x_2 + 2x_3 \end{bmatrix}$$

✅ Step 3: Multiply $x^T(Ax)$

$$x^T Ax = x_1(2x_1 - x_2) + x_2(-x_1 + 2x_2 - x_3) + x_3(-x_2 + 2x_3)$$

✅ Step 4: Expand

$$= 2x_1^2 - x_1x_2 - x_1x_2 + 2x_2^2 - x_2x_3 - x_2x_3 + 2x_3^2$$

$$= 2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_2 - 2x_2x_3$$

✅ Step 5: Rearrange (Important Trick)

Group terms:

$$= (x_1 - x_2)^2 + (x_2 - x_3)^2 + x_1^2 + x_3^2$$

✅ Step 6: Conclusion

- All terms are squares (≥ 0)
- Only zero when $x = 0$

👉 ✔ Therefore:

$$x^T Ax > 0 \quad \text{for all } x \neq 0$$

🎯 Final Answer

✔ Matrix A is Positive Definite

🔥 Quick Verification Trick (Exam Hack)

Try simple vectors:

Take $x = (1, 0, 0)$

$$x^T Ax = 2 > 0$$

Take $x = (1, 1, 1)$

$$= 2 + 2 + 2 - 2 - 2 = 2 > 0$$

✔ Always positive → confirms

💡 Important Insight

This matrix is actually a very common positive definite matrix (tridiagonal form used in engineering & numerical methods).

🧠 What You Should Learn From This

1. Expand $x^T Ax$ carefully
2. Try to convert into sum of squares
3. If everything is positive → ✔ positive definite

If you want next:

✔ Example of NOT positive definite (3×3)

✔ Example of semi-definite (where result = 0)

✔ Faster shortcut (Sylvester test for 3×3)

Just tell me 👍